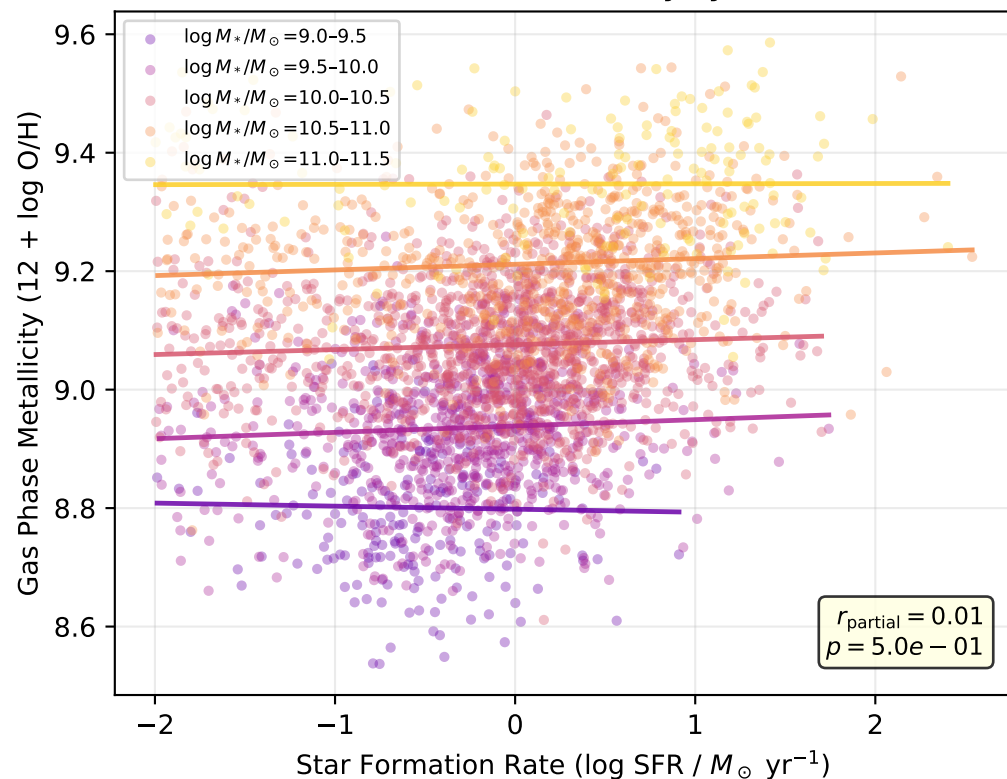


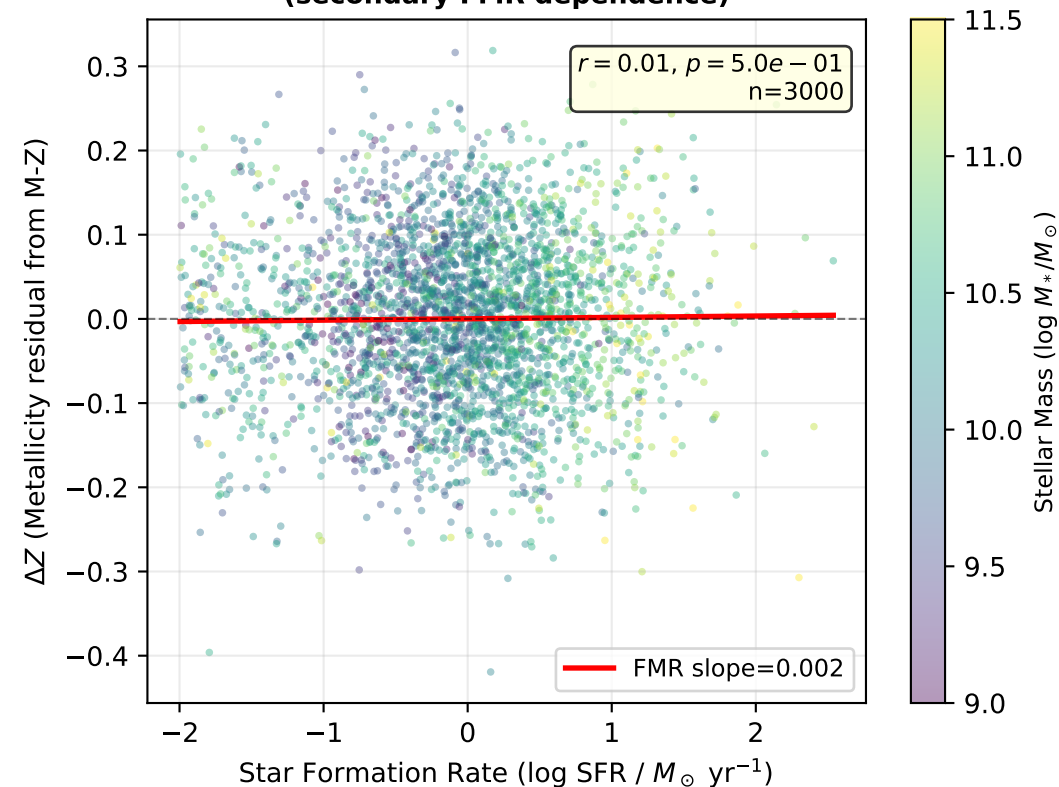
Test Case 6: Autonomous Discovery — SDSS Galaxy Causal Structure

ASTRA identified the Fundamental Metallicity Relation without prior knowledge

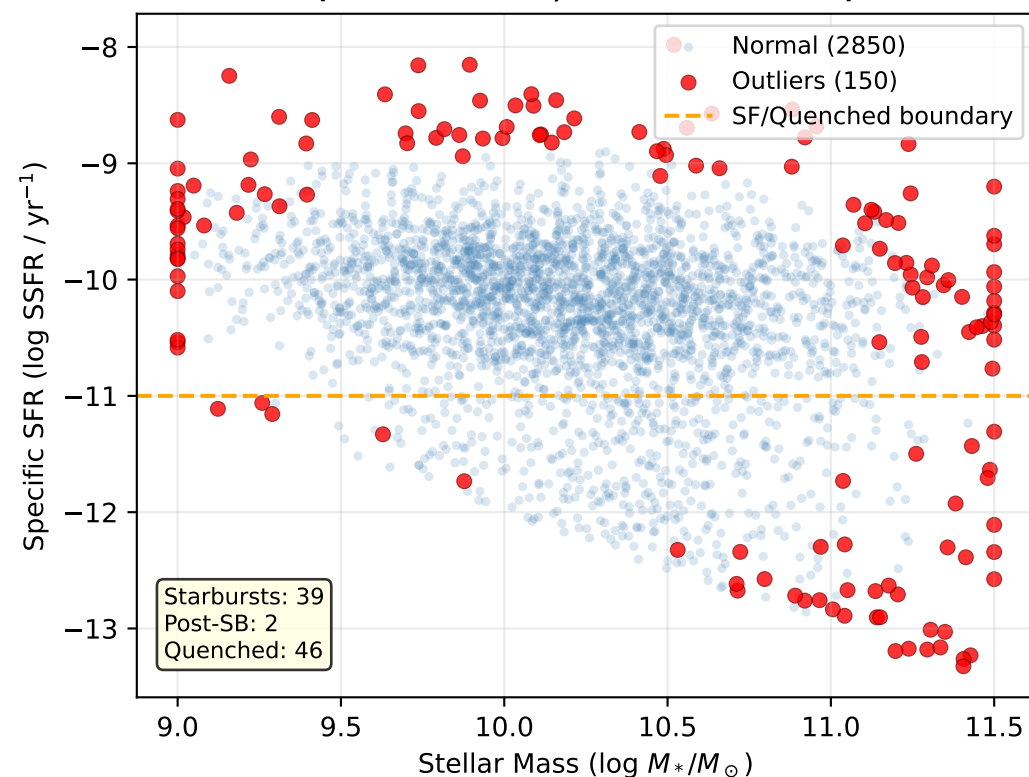
(a) Fundamental Metallicity Relation (discovered autonomously by ASTRA)



(b) SFR-driven Metallicity Scatter (secondary FMR dependence)



(c) Multi-dimensional Outlier Detection (IsolationForest, 5% contamination)



(d) Inferred Causal Structure (ASTRA autonomous discovery)

